Joe Miguel Robertazzi

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Education

Stanford University — Stanford, CA

Candidate for Bachelor of Science in Computer Science, Systems Track — *Sep. 2023 – June 2027*

GPA: 3.87/4.0

Previous/Ongoing Coursework: Parallel Computing, Digital System Design, Operating System Design, Computer System Architecture, Digital System Architecture, Computer Systems Programming, Data Structures & Algorithms, Circuits, Applied Matrix Theory, Differential Equations with Linear Algebra & Fourier Methods, Discrete Mathematics, Operating Systems, Probability, Proof Writing

Experience

Embedded Systems Intern — June 2025 – Aug. 2025

Stanford University – CS107E Course Infrastructure — *Stanford, CA*

- Developed and optimized low-level drivers for I2C, I2S, SPI, and DMA, integrating 12+ peripherals including RTCs, OLEDs, audio codecs, cameras, and motion sensors with improved performance and reliability
- Redesigned and stress-tested I2C & I2S APIs into a consistent hardware abstraction layer, improving robustness and simplifying integration across all devices
- Extended the MangoPi platform by adapting legacy projects to the new RISC-V architecture, designing custom PCBs for new assignments, and writing a guide to enable use of the Vector Extension in C/Assembly

Teaching Assistant — Apr. 2024 – Present

Stanford University – Department of Computer Science — *Stanford, CA*

- Conducted weekly 10+ student discussion sections of Stanford's introductory computer science courses CS106A/B
- Oversaw biweekly grading sessions with students to provide 1-on-1 feedback on all programming assignments
- Debugged code with students during office hours and graded midterm/final exams for both classes each quarter

Independent Student Researcher — Jan. 2021 – June 2023

University of Rochester — *Rochester, NY*

- Developed an independent research project investigating Earth's magnetic field and migratory bird patterns
- Analyzed 50M+ data points; created novel geographic visualizations using Python (Matplotlib, Seaborn, Pandas)
- Authored a formal research manuscript and presented at national competitions, including Regeneron STS

Projects

mangoDOOM - Bare-Metal DOOM Port | *C, ASM, RISC-V, MangoPi* — Jul. 2025 – Aug. 2025

- Ported the classic 1993 DOOM to a bare-metal RISC-V system (MangoPi), eliminating all OS dependencies by embedding game assets directly into the binary and replacing file I/O with direct memory access
- Developed bare-metal drivers for all core peripherals, including SPI for a 20+ fps LCD framebuffer, I2C for joystick/button input, and I2S with DMA for high-performance audio playback
- Implemented a custom C standard library from scratch, extending memory and string routines for bare-metal use

LEGOnardo DaVinci – LEGO Image Printer | *Python, C, ASM, RISC-V, MangoPi* — Jan. 2024 – Mar. 2024

- Designed and built a bare-metal image printer that recreates images using LEGO bricks as "ink"
- Developed image processing pipeline/UI in Python; implemented back-end control in C on a MangoPi (RISC-V)
- Integrated stepper motors and vacuum-based LEGO placement mechanism with custom drivers in C/Assembly
- Built a full user interface on MangoPi for image selection, scale, and real-time print progress tracking
- Delivered a working prototype that prints pixel-accurate LEGO images with a live feedback display

Research Project – Avian Migration Analysis | *Python, scikit-learn, Pandas, Seaborn* — Jan. 2020 – June 2023

- Created 100+ interactive geospatial maps using pygmt and Seaborn to visualize bird migration data
- Designed 4 modular analysis pipelines in Python with scikit-learn to evaluate environmental and magnetic impacts on bird migration
- Optimized data workflows for scalability and reproducibility; surfaced high-confidence correlations ($r > 0.7$) between magnetic fields and bird population decline

Technical Skills

Languages: C/C++, Assembly (RISC-V/MIPS), Python, HTML, CSS, TypeScript, Java

Developer Tools: Git, Linux, VS Code, PyCharm, LaTeX, Jupyter, Google Colab, MangoPi, Raspberry Pi

Libraries: pandas, NumPy, scikit-learn, Matplotlib, Seaborn